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Chips: The Biggest AI Bottleneck

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IMPORTANT RESEARCH DISCLOSURES LOCATED IN APPENDIX 1.

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Key points

1. Today's AI race is inherently a chip race. No country controls the overall chip supply chain in the short to medium term.

2. Key bottlenecks lie in advanced chips and specialized equipment, concentrated mainly in Taiwan and the Netherlands.

3. The US leads in advanced compute, but is constrained by its reliance on Taiwan's TSMC for chip manufacturing.

4. China's AI chips perform ~5x below US alternatives — a gap that could widen to 17x — though it is expanding in legacy node production. China maintains access to ~2.8mn H100-equivalent GPUs partly via remote cloud.

5. No single country can achieve full chip supply chain independence in the near term. The EU is on track for only 12% of global production by 2030, short of its 20% target.

6. The 2025 export control escalation — US chip controls alongside China's rare earth restrictions — illustrates how semiconductors and critical minerals have become interlinked economic tools.

7. The chip industry will bifurcate between AI leaders and laggards beyond 2026, impacting the ongoing memory shortage. Software efficiency gains, such as Google's TurboQuant, can rapidly reprice the hardware stack.

Key facts

~\$1.5trn	in global semiconductor market revenue by 2030. AI will drive most of this.
60%	of global chip production was US-made in the 1970s, vs 11% by 2019.
40%	of Taiwan's chip supply chains should be reshored to the US, per Commerce Secretary Lutnick.
15 years	for the US to build a sustainable domestic chip manufacturing industry and packaging industry.
90%	of the world's most advanced semiconductors are produced in Taiwan.
11%	of US GDP is at risk of a Taiwan Strait disruption (SIA). A disruption could cost \$5trn in year one (Bloomberg).
~80%	Chinese chips reach of Nvidia's A100 performance, but this gap is projected to widen significantly as US chips advance.
7 months	Chinese LLMS lag behind US counterparts.
2027	China's mature node capacity rises to 47% (vs. 34% in 2024). Taiwan's share would decline from 43% to 36%.
53%	of China's semiconductor exports were memory chips in Dec 2025.



1. For the 2030 projections, see Company, Manufacturing. “2026 Technology Symposium Highlights.” Tsmc.com. Taiwan Semiconductor Manufacturing Company. https://www.tsmc.com/english/symposium_highlights/2026.
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3. Dylan Butts, “‘Impossible’: Taiwan Pushes Back against Washington’s 40% Chip Supply Relocation Goal,” CNBC, February 10, 2026, <https://www.cnbc.com/2026/02/10/taiwan-chips-us-supply-chain-lutnick-trade-deal.html>.
4. Deutsche Bank Research analyst projections.
5. Aqib Zakaria, “How Much Compute Does China Have?” Chinatalk, April 7, 2026, <https://www.chinatalk.media/p/how-many-chips-does-china-have>.
6. “Special Report 12/2025: The EU’s Strategy for Microchips – Reasonable Progress in Its Implementation but the Chips Act Is Very Unlikely to Be Sufficient to Reach the Overly Ambitious Digital Decade Target,” (Luxembourg: European Court of Auditors, 2025).
7. Deutsche Bank Research.
8. Chris Anstey, “Bloomberg New Economy: China ‘Blockade Simulation’ Raises \$5 Trillion Risk,” Bloomberg.com, May 25, 2024, <https://www.bloomberg.com/news/newsletters/2024-05-25/bloomberg-new-economy-china-blockade-simulation-raises-5-trillion-risk>. For SIA figure, see Tripp Mickle, “The Looming Taiwan Chip Disaster That Silicon Valley Has Long Ignored,” Feb 24, 2026, <https://www.nytimes.com/2026/02/24/technology/taiwan-china-chips-silicon-valley-tsmc.html>
9. Chris McGuire, “China’s AI Chip Deficit: Why Huawei Can’t Catch Nvidia and U.S. Export Controls Should Remain,” Council on Foreign Relations, December 15, 2025, <https://www.cfr.org/articles/chinas-ai-chip-deficit-why-huawei-cant-catch-nvidia-and-us-export-controls-should-remain>.
10. Luke Emberson, “Chinese AI Models Have Lagged the US Frontier by 7 Months on Average since 2023,” Epoch AI, 2023, <https://epoch.ai/data-insights/us-vs-china-eci/>.
11. “Chapter 9: Chained to China,” U.S.-China Economic and Security Review Commission.
12. General Administration of Customs, People’s Republic of China.



Appendix 1

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